Room-temperature bonded thermally conductive semiconductor interfaces

16

Zhe Cheng[a] *and Samuel Graham*[b]
[a]Department of Materials Science and Engineering, University of Illinois at Urbana-Champaign, Urbana, IL, United States, [b]George W. Woodruff School of Mechanical Engineering, Georgia Institute of Technology, Atlanta, GA, United States

16.1 Introduction

GaN has great potential to be widely used for power and radio-frequency (RF) electronics, especially for high-power and high-frequency applications [1,2]. The technical innovations brought by GaN-based high electron mobility transistors (HEMTs) advance the development of areas including defense (radar systems), energy infrastructure, self-driving/wireless charging vehicles, and wireless communications [2–4], leading to size-compact, fast response, and energy efficiency applications. The amount of GaN needed to isolate a certain voltage is 10 times less than that of Si, resulting in size-compact devices [2,3]. The switching speed of GaN devices can be 10 times higher than Si counterparts, facilitating applications which require short response times such as self-driving cars [2]. About 5%–10% of the world's electricity is wasted as dissipated heat in the power electronics circuits in which GaN technology can help reduce the amount of waste heat [5].

For GaN HEMTs, localized Joule heating due to the large voltage difference during switching near the gate increases the local temperature significantly, which degrades the stability and performance and lowers the lifetime of the devices [6]. The local heat flux of a GaN HEMT can reach more than 10 times larger than that of the sun surface [7]. Moreover, the channel temperature is strongly related to the device lifetime. 25 K decrease in channel temperature results in about 10 times increase in device lifetimes for GaN-on-SiC devices [8]. This highlights the importance of thermal characterizations of related materials and interfaces and proper thermal management of related devices.

Integration of GaN or other wide/ultrawide bandgap semiconductor devices with high thermal conductivity substrates such as SiC or diamond is a typical technique to extract heat from these devices. The thermal boundary conductance (TBC) between devices and the substrates are the key for proper thermal management since the thermal resistances from the interfaces account for a large or even dominant portion of the total thermal resistance. Understanding the TBC between the devices and substrates are important for both fundamentally thermal science and real-world applications.

Thermal Management of Gallium Nitride Electronics. https://doi.org/10.1016/B978-0-12-821084-0.00008-1
Copyright © 2022 Elsevier Ltd. All rights reserved.

This work will provide a brief review of thermally conductive semiconductor interfaces bonded by a room-temperature bonding technique (surface-activated bonding technique). GaN and Ga_2O_3 are integrated with high thermal conductivity substrates (SiC and diamond). The thermal conductivity of bulk GaN and thin films is also summarized.

16.2 Thermal measurement techniques

Commonly used thermal characterization techniques include time-domain thermoreflectance (TDTR), 3-Omega, frequency-domain thermoreflectance (FDTR), laser-flash technique, and steady-state heat flow. TDTR has been widely used for characterizing thermal transport in bulk materials, micro/nano films, and across interfaces [6,9,10]. FDTR is a modified version of TDTR by fixing the delay time and variating the modulation frequency [11]. The other techniques are very limited for characterizing TBC.

TDTR is an ultrafast-laser-based optical pump-probe technique. A femtosecond pulse laser is split into two beams: a pump beam and a probe beam. The pump beam is chopped by an electro-optical modulator (EOM) so that the laser beam heats the to-be-measured sample surface periodically. The sample surface is usually coated with a layer of metal such as Al as transducer. The probe beam detects the temperature variation of the sample surface via thermoreflectance. By using a moving mechanical stage, the probe beam is delayed with a certain time, which enables the probe beam to detect the surface temperature with a delay time after the surface heating by the pump beam. After being picked up by a photodiode and a lock-in amplifier, the measured signal is compared with an analytical heat transfer solution of the sample structure to fit for the unknown thermal parameters. All the above-mentioned techniques can be used to measure thermal conductivity. But for TBC, TDTR is the most powerful tool. Most of the TBC data in the literature are measured by TDTR [12–14].

16.3 Thermal conductivity of bulk GaN and thin films

For the intrinsic thermal conductivity of bulk GaN materials, Table 16.1 summarizes the calculated and measured values at room temperature. Slack estimated GaN thermal conductivity as 170 W/m-K at 1973, while the first measurement performed at 1977 showed a lower value (130 W/m-K). The intrinsic thermal conductivity of GaN was theoretically predicted very recently after the first-principle density-function-theory (DFT) calculation was developed [16]. The original calculation only considers three phonon scatterings when solving Boltzmann transport equation (BTE). After recent development of four-phonon scattering calculation, the prediction of thermal conductivity of perfect GaN single crystal becomes more accurate and the thermal conductivity was reduced from 240 to 230 W/m-K [15,16]. An interesting finding obtained by theoretical calculation is that the gallium isotope scattering accounts for a large part of the phonon scatterings which limits the thermal conductivity of GaN. Thus, for gallium-isotope-pure

Table 16.1 A historical summary of thermal conductivity of bulk GaN at room temperature.

References	Year	Thermal conductivity (W/m-K)	Techniques/methods
Yang [15]	2019	230 (372 Isotope-pure)	First-principle (3+4 phonon)
Lindsay [16]	2012	240 (393 Isotope-pure)	First-principle (3 phonon)
Slack [17]	1973	170	Estimation
Li [18]	2020	216	TDTR
Zheng [19]	2019	195	TDTR
Rounds [20]	2018	224/211/196/164	3-Omega
Paskov [21]	2017	245	3-Omega
Jezowski [22]	2015	269	Steady-state Heat Flow
Simon [23]	2014	162/205/225	3-Omega
Shibata [24]	2007	252	Laser flash
Mion [25]	2006	230	3-Omega
Jezowski [26]	2003	160/188/226	Steady-state Heat Flow
Slack [27]	2002	227	Steady-state Heat Flow
Sichel [28]	1977	130	Steady-state Heat Flow

GaN, the predicted thermal conductivity is as high as 393 W/m-K (considering 3-phonon scattering) and 372 W/m-K (considering both 3 and 4 phonon scatterings) [15,16]. Zheng et al. measured the thermal conductivity of gallium-isotope-pure GaN and only found a ~15% increase in thermal conductivity [19]. The challenge to obtain high thermal conductivity GaN is the difficulty to obtain high-purity isotope-pure gallium and high-quality GaN [19]. This topic is worth further exploration since the high thermal conductivity of isotope-pure GaN facilitates heat dissipation of GaN-based electronics significantly and isotope compositions do not affect electrical properties of GaN theoretically.

For experimental measurements of bulk GaN thermal conductivity, Sichel et al. performed the measurements by the steady-state heat flux technique decades ago [28]. With the development of the measurement techniques and GaN growth techniques, different thermal conductivity values have been reported [18–27]. Most of the measured thermal conductivity are lower than 230 W/m-K because of structural imperfections such as impurities, dislocations, and vacancies [18]. The reported thermal conductivity values higher than 230 W/m-K are due to thermal measurement inaccuracy [19]. Different growth techniques and different growth conditions result in different qualities of GaN materials, correspondingly different thermal conductivity. Due to the importance of GaN electronics, there are still large space for studies of correlating these important parameters (growth techniques and growth conditions) with the GaN thermal conductivity. How structural imperfections such as doping and dislocation density affect the thermal conductivity of bulk GaN needs further study [29].

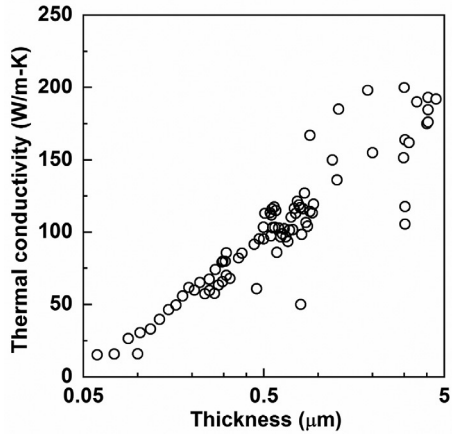


Fig. 16.1 A review of experimentally measured thermal conductivity of GaN thin films with different thicknesses [18,29–41].

For GaN micro/nano structures that compose related devices, their thermal conductivities are reduced compared with the counterpart of bulk GaN materials [29,32,42]. GaN devices are expected to be hotter than those estimated based on the bulk GaN thermal conductivity. Fig. 16.1 shows a review of experimentally measured thickness-dependent thermal conductivity of GaN thin films [18,29–41]. We observe strong size effect on these measured thermal conductivity that increases with increasing film thickness. When the thickness reaches several micrometers, the measured thermal conductivity approaches the thermal conductivity of bulk GaN. In bulk GaN, phonon–phonon scatterings are the dominant mechanism to determine the phonon mean free path at room temperature. While in thin films, the film boundaries are additional scattering sources to scatter phonons and limit phonon mean free path. Short phonon mean free path leads to reduced thermal conductivity. The thinner the films are, the shorter the phonon mean free path are, and correspondingly the lower the thermal conductivity are. The GaN thin films not only have lower thermal conductivity than the bulk GaN but also bring in more interfaces that introduce thermal boundary resistances.

16.4 A summary of TBC of GaN–SiC and GaN–diamond interfaces

The thermal conductivity of single-crystal SiC and diamond are 380 and >2000 W/m-K, much higher than that of GaN [31]. When GaN-based power electronics is heterogeneously integrated with SiC and diamond [1,31,42,43], there exists a trade-off between the advantage of high thermal conductivity substrates and the additional thermal resistance of interfaces brought by heterogeneous integration. The thermal resistances of GaN–SiC interfaces and GaN–diamond interfaces account for a large part of the total thermal resistances from the hotspots to the heat sink [44]. It is a great challenge to achieve high TBC for these interfaces.

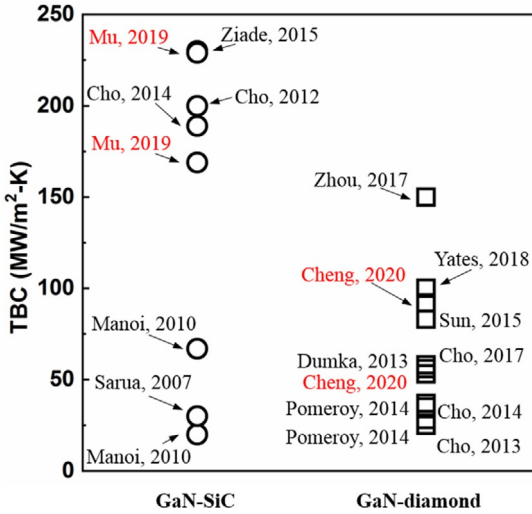


Fig. 16.2 A review of TBC of GaN-related interfaces for GaN-on-SiC and GaN-on-diamond Devices: GaN–SiC [34,35,42,43,45,46] and GaN–diamond [31,44,47–53]. The TBC of the surface-activated bonded interfaces are marked *red*.

Fig. 16.2 shows a review of experimentally measured TBC of GaN–SiC and GaN–diamond interfaces [31,34,35,42–53]. Great efforts have been paid to improve the thermal transport across these technologically important interfaces. GaN can be grown on SiC directly or with an AlN transition layer. The direct growth of GaN on SiC leads to a relatively lower GaN quality due to the lattice mismatch. But with an AlN transition layer, it introduces additional thermal resistances including the thermal resistance of the AlN layer and interfaces. Recent work showed that high TBC and high-quality GaN can be achieved by bonding GaN with SiC directly at room temperature with the surface-activated bonding technique [42]. After annealing, the TBC of the bonded GaN–SiC interfaces can be 229 MW/m^2-K (marked red in the figure), almost the same as the TBC of directly grown GaN–SiC interfaces [42,43].

For GaN–diamond interfaces, high-quality GaN cannot be directly grown on single-crystal diamond due to the large lattice mismatch between GaN and diamond. Thus, chemical vapor deposition (CVD) of diamond is used to grow polycrystalline diamond on GaN with a thin protective layer such as AlN [52]. This protective layer introduces additional thermal resistance between the GaN devices and the diamond layer. Furthermore, the diamond near the growth interface are nanocrystalline [54]. Due to the strong phonon-grain boundary scatterings and phonon-defect scatterings, the thermal conductivity of the nanocrystalline diamond is significantly reduced compared with that of single crystal diamond, which is only tens of W/m-K [1,55]. The crystal size of the diamond increases as the growth of CVD diamond. The crystal size does not increase to tens of micrometers until the thickness of the diamond film reaches tens of micrometers [6]. It is a technical challenge to bring high thermal conductivity diamond close to the hotspots in GaN with high TBC. Recent work showed that this can be achieved by surface-activated bonding of GaN with single crystal

diamond [31]. Single crystal high thermal conductivity diamond can be integrated with GaN by surface-activated bonding at room temperature, achieving low residual stress for the bonded interfaces. The TBC is among the highest values reported for GaN–diamond interfaces [31]. The high TBC of the GaN–diamond interfaces and high thermal conductivity of single crystal diamond approached the up limit of the best cooling performance for any materials. But the limitation of this bonded GaN–diamond technique is the large-area wafer-scale polishing and bonding of single crystal diamond, which needs further study for scale-up production.

16.5 Surface-activated bonding technique

Surface-activated bonding technique is a bonding method which enables wafer-scale bonding at room temperature [30,31]. The to-be-bonded surfaces which need to be ultra-smooth (root-mean-square roughness <1 nm) are usually polished by chemical–mechanical polishing process before bonding. Rough surfaces or dusts result in unbonded area or voids at the interface. Fig. 16.3 shows the schematic diagram of the bonding process of GaN with SiC substrates. The GaN-on-sapphire wafer and the SiC wafer after polishing are transferred to a vacuum chamber with ultra-high vacuum ($\sim 5 \times 10^{-6}$ Pa). Ar ion beams are used to activate both surfaces to remove any contaminations and oxides, and activate the surfaces. When activating SiC, the SiC surface is usually carbon-rich. Therefore, in the Ar ion source, a Si source is added to form a mixture beam of Ar and Si. The Si-containing Ar ion beam adds Si elements at the interface, which facilitate strong bonding of the GaN–SiC interfaces. The acceleration voltage and current of the Ar ion beam are typically 1.2 kV and 400 mA, respectively. The acceleration voltage and current of the Si-containing Ar ion source are typically 1.0 kV and 100 mA, respectively. The surface activation leaves dangling bonds at the surface. The two activated wafers are pressed together with a pressure of 5 MPa at room temperature for 300 s. Covalent bonds form at the interfaces and the wafers are bonded together. The sapphire is removed by laser lift off with a wavelength of 248 nm. A thin layer of GaN (about 2 μm) is left on the SiC and well-bonded.

GaN can be bonded with SiC directly without any interfacial layer while GaN is very difficult to bond with diamond due to chemical challenges since gallium element and nitrogen element are difficult to form covalent bonds with carbon element. The conventional surface-activated bonding technique needs to be modified to bond GaN with single crystal diamond. Fig. 16.4 shows the schematic diagram of three different bonding methods. Fig. 16.4A is the conventional surface-activated bonding technique and Fig. 16.4B is the conventional surface-activated bonding technique with Si-containing Ar ion beams. Fig. 16.4C is the modified surface-activated bonding technique with deposited Si. Si element can form strong covalent bonds with carbon element. Therefore, Si elements are added at the interfaces to obtain strong and uniform GaN–diamond interfaces. The Si-containing Ar ion beams result in a thin layer of Si at the interface while the sputtering-deposited Si at the interface is much thicker.

β-Ga_2O_3 is an ultrawide bandgap semiconductor, which has a thermal conductivity order of magnitude lower than GaN [56]. High thermal conductivity substrate

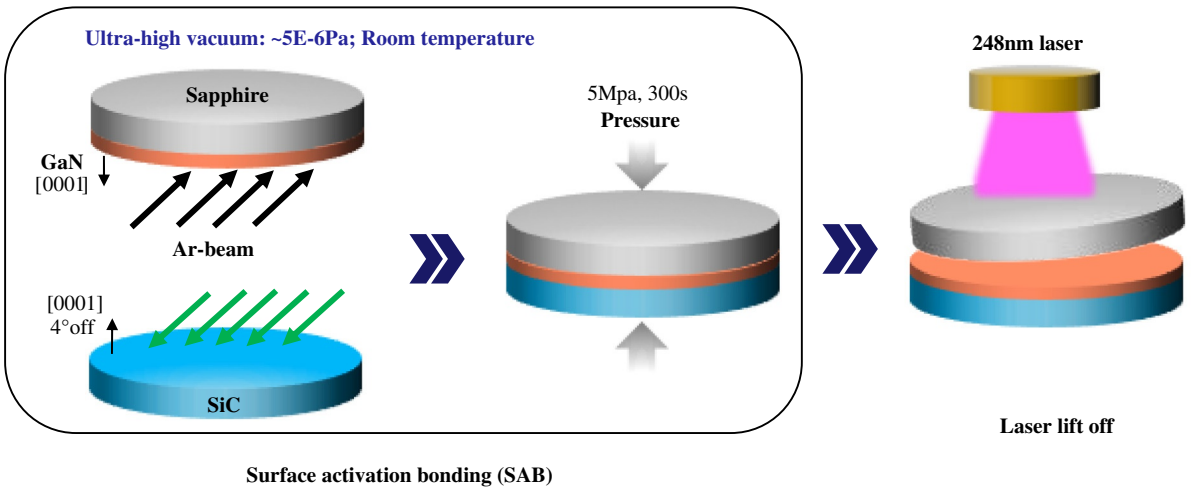


Fig. 16.3 Schematic diagram of the surface-activated bonding of GaN with SiC [30].
Reproduced with permission from F. Mu, et al., High thermal boundary conductance across bonded heterogeneous GaN–SiC interfaces, ACS Appl. Mater. Interfaces 11 (36) (2019) 33428–33434.

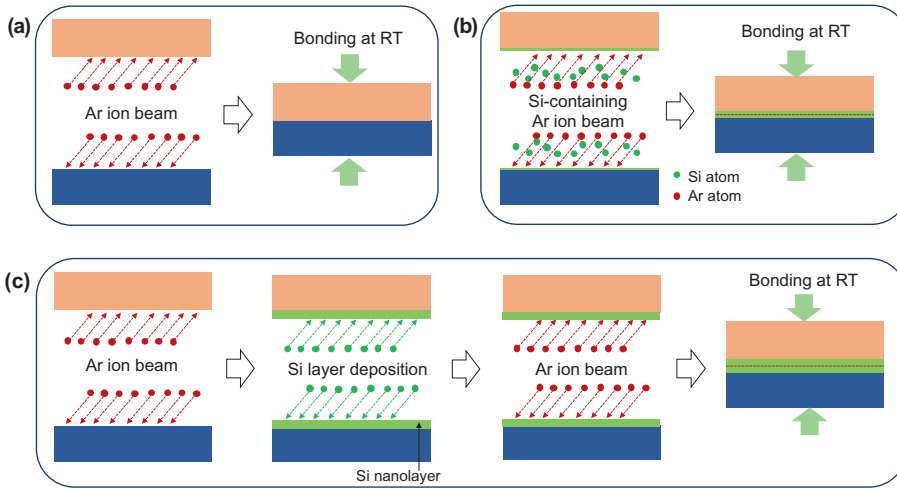


Fig. 16.4 Comparison of different surface-activated bonding techniques: (A) conventional surface-activated bonding technique; (B) conventional surface-activated bonding technique with Si-containing Ar ion beam; (C) modified surface-activated bonding technique with sputtered Si layer [31].
Reproduced with permission from Z. Cheng, F. Mu, L. Yates, T. Suga, S. Graham, Interfacial thermal conductance across room-temperature-bonded GaN/diamond interfaces for GaN-on-diamond devices, ACS Appl. Mater. Interfaces 12 (2020) 8376–8384.

facilitates heat dissipation for β-Ga_2O_3 based devices. However, epitaxial growth of β-Ga_2O_3 thin films on heterogeneous substrates are difficult. Therefore, smart-cut technique is promising to cut nanoscale β-Ga_2O_3 monocrystalline thin film. The high-quality nanoscale β-Ga_2O_3 thin films enable further growth of β-Ga_2O_3 devices with excellent heat dissipation [57].

Fig. 16.5 shows the schematic diagram of smart-cut β-Ga_2O_3 thin films bonded with SiC substrates [57]. After the β-Ga_2O_3 and SiC wafers are polished, the surface roughness are about 0.3 nm. Hydrogen ions are implanted into the β-Ga_2O_3 wafer with an energy of 35 keV and implantation dose of around 1×10^{17} cm^{-2} at room temperature. The ion implantation is performed with a 7-degree tilt to minimize the ion channeling effect. The implanted hydrogen ions accumulate in the β-Ga_2O_3 wafer at 200–400 nm beneath the surface. Al_2O_3 is deposited on the surface for surface-activated bonding. This Al_2O_3 can produce uniform bonding and block possible electrical breakdown at the defective interface [57]. During the surface activation of SiC, a silicon source is added into the Ar ion source to produce a Si–Ar ion mixture. The added Si ions is used to prevent the SiC surface from carbonation. The Ar ions break the chemical bonds of the Al_2O_3 and SiC surface to create dangling bonds. After pressing the two wafers together, covalent bonds form at the interface. The Si-containing Ar ion source has a voltage and current of 1.0 kV and 100 mA. The chamber pressure during surface activation is 0.4 Pa [57].

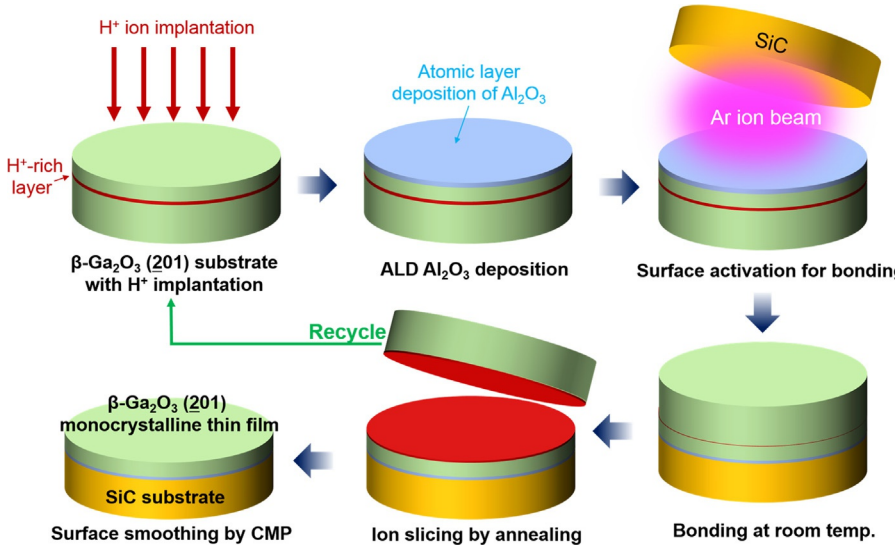


Fig. 16.5 Schematic diagram of bonding nanoscale β-Ga_2O_3 monocrystalline thin films with SiC wafers. After the β-Ga_2O_3 wafer is implanted by H ions, a layer of Al_2O_3 is deposited on the surface. Then, the Al_2O_3 coated β-Ga_2O_3 wafer is bonded with a 4H-SiC wafer at room temperature. After heating up to 450 °C, a thin layer of monocrystalline β-Ga_2O_3 is exfoliated and transferred on the SiC wafer. After the surface is polished by chemical mechanical polishing, the rest part of the β-Ga_2O_3 wafer can be used repeatedly [57].
Reproduced with permission from Z. Cheng, et al., Thermal transport across ion-cut monocrystalline β-Ga_2O_3 thin films and bonded β-Ga_2O_3–SiC interfaces, ACS Appl. Mater. Interfaces 12 (40) (2020) 44943–44951

To exfoliate the thin β-Ga_2O_3 layer from the β-Ga_2O_3 wafer, the bonded wafers are heated up to 450 °C. The formed hydrogen splits a thin layer of β-Ga_2O_3 (<400 nm) from the β-Ga_2O_3 wafer. The β-Ga_2O_3 thin film transferred on the SiC wafer has a rough surface. The β-Ga_2O_3 layer is polished by chemical mechanical polishing technique to obtain smooth surface. The thickness of the β-Ga_2O_3 thin films can be well-controlled by the polishing process from 300 nm to less than 100 nm.

16.6 Thermal conductance of bonded interfaces

The thermal properties of these bonded interfaces are measured by TDTR. Low modulation frequencies (2.2 or 3.6 MHz) are usually used to obtain a large thermal penetration depth for large sensitivity of the TBC of buried bonded interfaces. Fig. 16.6A shows the bonded GaN-SiC wafers [30]. The sapphire substrate has been removed from the GaN layer by laser lift off process. The GaN layer (about 2 μm thick) can be polished to thinner films for epitaxial growth. Here, the thickness of the GaN film was polished for optimization of TDTR measurements. Multiple spots on the sample with different GaN thickness were measured. A piece of the sample was annealed in

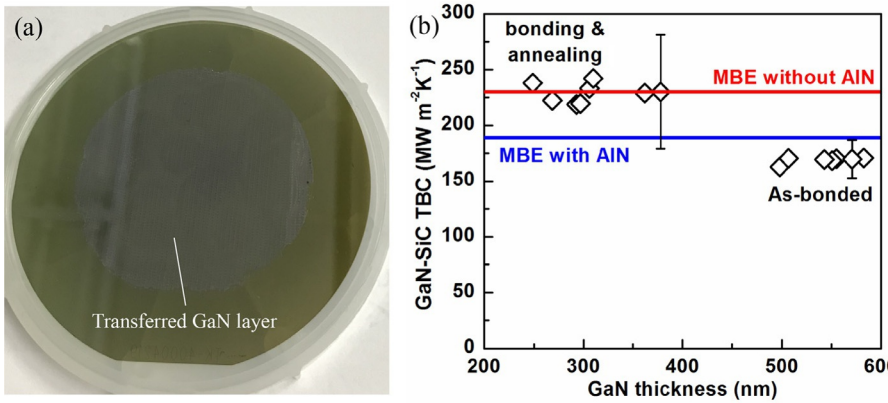


Fig. 16.6 (A) Bonded GaN–SiC wafers. (B) GaN–SiC TBC for annealed and as-bonded samples [30]. The TBC of MBE-grown GaN on SiC with and without a AlN layer are from literature [33,43].
Reproduced with permission from F. Mu, et al., High thermal boundary conductance across bonded heterogeneous GaN–SiC interfaces, ACS Appl. Mater. Interfaces 11 (36) (2019) 33428–33434.

N_2 at 1273 K. The measured TBC of the as-bonded and annealed samples as a function of GaN thickness are shown in Fig. 16.6B [30]. The TBC of epitaxial grown GaN–SiC interfaces are also included for comparison [33,43].

The measured TBC is independent on the GaN thickness. The TBC of the as-bonded GaN–SiC interfaces is lower than the TBC of the epitaxial grown GaN–SiC interfaces with or without AlN transition layers. However, the annealing at 1273 K improves the interfacial thermal conductance of the bonded GaN–SiC interfaces and achieves high thermal conductance. The measured TBC is almost the same as the epitaxial GaN–SiC interfaces without AlN transition layers. The bonded interfaces remove the low-quality GaN near the nucleation interface and achieve high TBC, which facilitate heat dissipation of GaN-based electronics.

To understand the improvement of interfacial thermal conductance of the bonded interfaces after annealing at high temperatures, a scanning transmission electron microscopy was used to study the cross sections of the bonded interfaces before and after annealing, as shown in Fig. 16.7. Fig. 16.7A and B are the bright-field image and the high-angle annular dark-field image of as-bonded GaN–SiC interfaces while Fig. 16.7C and D are the bright-field image and the high-angle annular dark-field image of annealed GaN–SiC interfaces. Due to the ion implantation, the interface has a layer of amorphous layer with a thickness of 3 nm. After annealing at high temperature, the amorphous layer crystalizes. Amorphous layers have low thermal conductivity than their crystalline counterparts. This explains the improvement of the TBC of bond GaN–SiC interfaces after annealing (Fig. 16.8).

To further understanding the effect of annealing on the bonded interfaces, electron energy loss spectroscopy is used to map the element distribution of the interfaces.

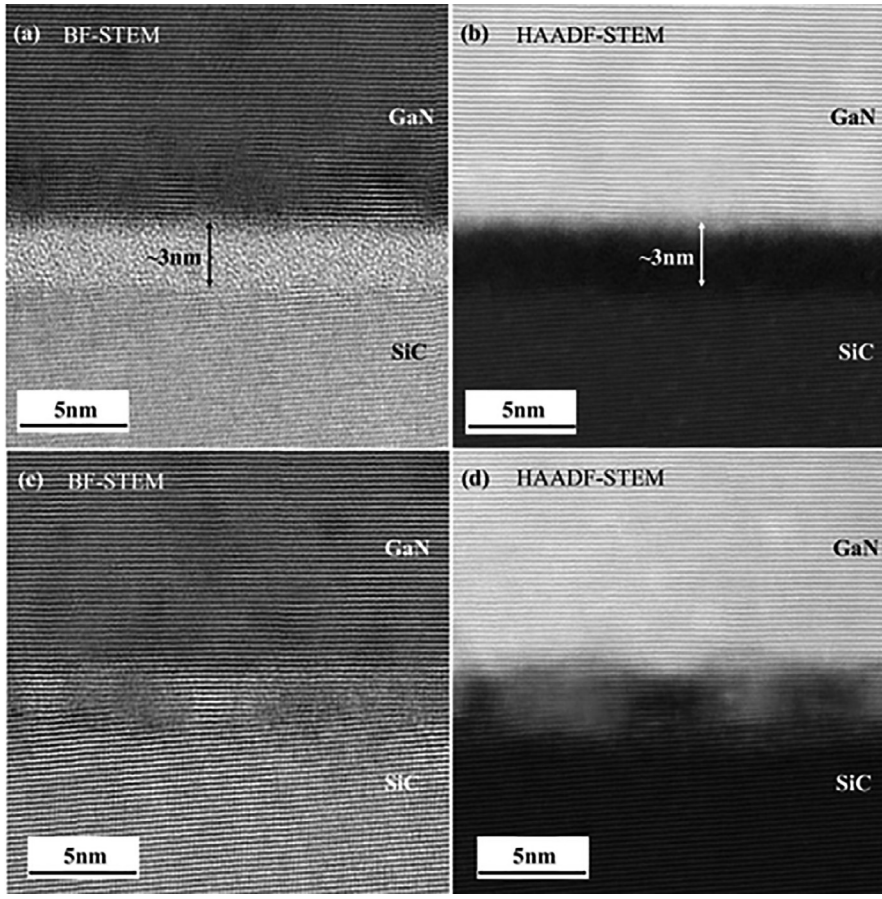


Fig. 16.7 The effect of annealing on the bonded GaN–SiC interfaces. Cross-section high-resolution scanning transmission electron microscopy images of GaN–SiC interface bonded at room temperature and after annealing at 1273 K: (A) bright-field image of as-bonded GaN–SiC interfaces; (B) high-angle annular dark-field image of as-bonded GaN–SiC interfaces; (C) bright-field image of annealed GaN–SiC interfaces; (D) high-angle annular dark-field image of annealed GaN–SiC interfaces [30].
Reproduced with permission from F. Mu, et al., High thermal boundary conductance across bonded heterogeneous GaN–SiC interfaces, ACS Appl. Mater. Interfaces 11 (36) (2019) 33428–33434.

The majority of the amorphous layer is amorphous SiC. Interfacial diffusions of Ga, N, Si, and C elements are observed at the interface after annealing. For the as-bonded GaN–SiC interface, Ar is trapped at the interface after bonding and most of them exists at the SiC side. After annealing, the distribution of Ar elements redistribute to clusters and most of the interfacial area is free of Ar. This redistribution of Ar should affect TBC as well even though the detailed mechanisms need further study to figure out.

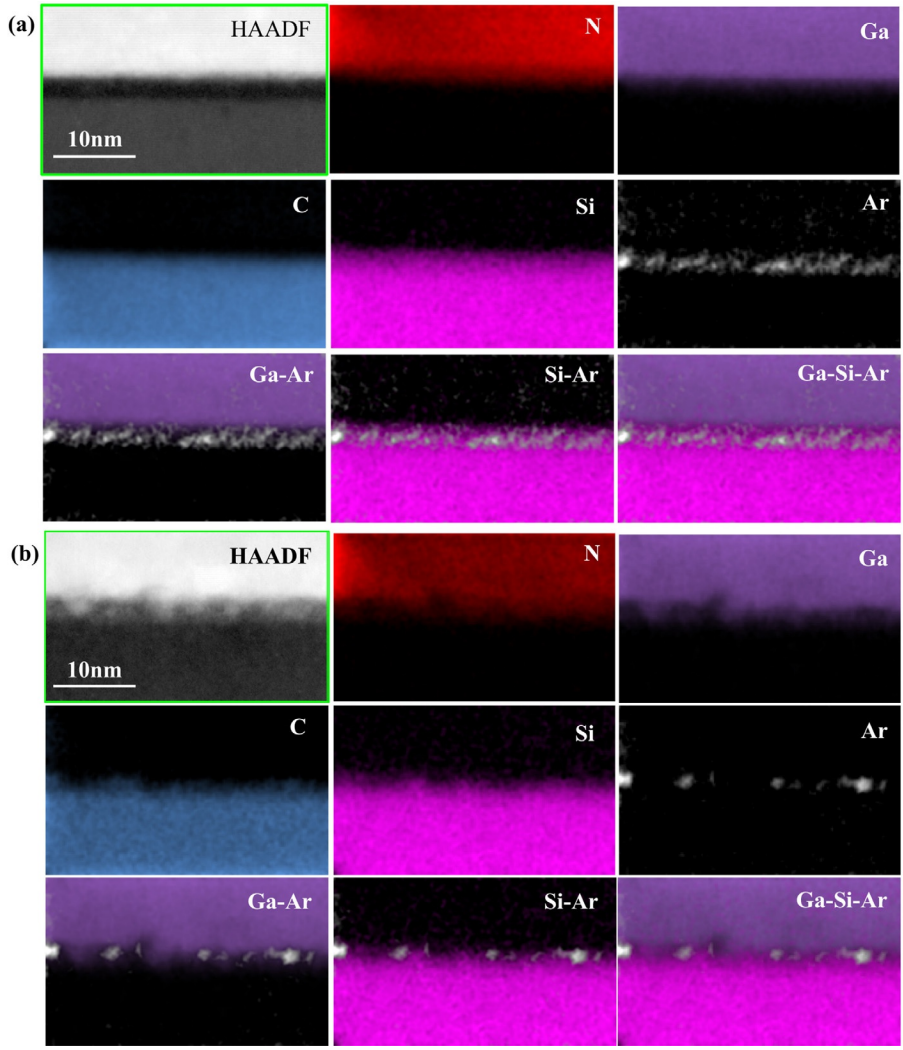


Fig. 16.8 Electron energy loss spectroscopy mapping of GaN-SiC bonded interfaces before and after annealing at N_2: (A) at room temperature and (B) after annealing at 1273 K. The N, Ga, C, Si, and Ar maps are marked in *red*, *purple*, *blue*, *pink*, and *white*, respectively [30]. Reproduced with permission from F. Mu, et al., High thermal boundary conductance across bonded heterogeneous GaN–SiC interfaces, ACS Appl. Mater. Interfaces 11 (36) (2019) 33428–33434.

For current GaN-on-diamond devices, GaN devices are fabricated on silicon substrates first. Then the silicon substrates are etched away. After depositing a thin protective layer such as AlN on GaN, chemical vapor deposited (CVD) diamond is grown on the GaN [6,52,53,58]. However, the diamond near the nucleation interface is nanocrystalline, which has a thermal conductivity orders of magnitude lower than bulk

diamond [1,54,59]. Moreover, the protective layer introduces additional thermal resistances even though it protects the GaN materials from plasma contamination during CVD diamond growth [52,53]. Unlike conventional bonding techniques that need to be performed at high temperatures, surface-activated bonding technique can bond GaN with single crystal diamond at room temperature, which avoids the thermal stress after bonding when cooling down to room temperature.

Fig. 16.9A shows the cross-section image of the bonded GaN–diamond interface with deposited Si. The deposited Si is about 5 nm thick on each side of the bonded interface. A 3-nm-thick amorphous diamond shows up due to the ion implantation during surface activation. Fig. 16.9B shows the cross-section image of the bonded GaN–diamond interface with Si-containing Ar ion beams. Only a thin layer of Si (about 4 nm thick) shows up at the interface. The thin interfacial layer facilitates thermal transport. As shown in Fig. 16.9C, the TBC of the bonded GaN–diamond interface with Si-containing Ar ion beams has a TBC of 90 MW/m^2-K which is one of the highest TBC measured for GaN–diamond interfaces and much higher than that with deposited Si [31]. Very weak temperature dependence of the measured TBC is observed, which is possibly due to the complicated interfacial structure.

Compared to the current GaN-on-diamond devices, the integration of GaN with single crystal diamond by surface-activated bonding technique directly brings high thermal conductivity diamond close to the heating source. The thermal conductivity of single crystal diamond is orders of magnitude higher than the nanoscrystalline diamond. Furthermore, the direct bonding of GaN with diamond at room temperature creates high TBC with low thermal stress. This avoids the problems of stress-induced wafer bowing [4,7,60,61]. The room-temperature bonding of GaN with single crystal diamond may provide a new way to fabricate GaN-on-diamond devices with both excellent thermal and mechanical performance.

To obtain an estimate how the GaN–substrate TBC affect device temperature, an analytical heat transfer model is used to calculate the peak temperature of an 800-nm GaN device with 10 fingers under fully open channel condition [31]. The dimensions of the device is show in Fig. 16.10A [31]. The used power density is 10 W/mm, and the heating source width is 4 μm. The heating source length is 500 μm, and the gate–gate spacing is 50 μm. In the calculation, the thermal conductivities of SiC and diamond are 380 and 2000 W/m-K [19,63].

Fig. 16.10B shows the effect of the GaN–substrate TBC on the peak device temperature [31]. In the studied TBC range, the peak temperature of GaN-on-diamond is much lower than that of the GaN-on-SiC. The measured TBC of the bonded GaN–diamond interfaces are marked as gray circles on the figure. The peak device temperature with the measured TBC of the bonded GaN–diamond interfaces are approaching the lowest temperature (infinite high TBC). The further increase in the GaN–diamond TBC will decrease the device maximum temperature but not much.

Additionally, surface-activated bonding technique is used to bonded β-Ga_2O_3 wafers and SiC wafers for smart-cut processing [57]. Fig. 16.11A shows the image of a 2-in. β-Ga_2O_3 wafer bonded with a 4-in. 4H-SiC wafer [57]. A layer of Al_2O_3 is added at the interface to obtain uniform bonding even though direct

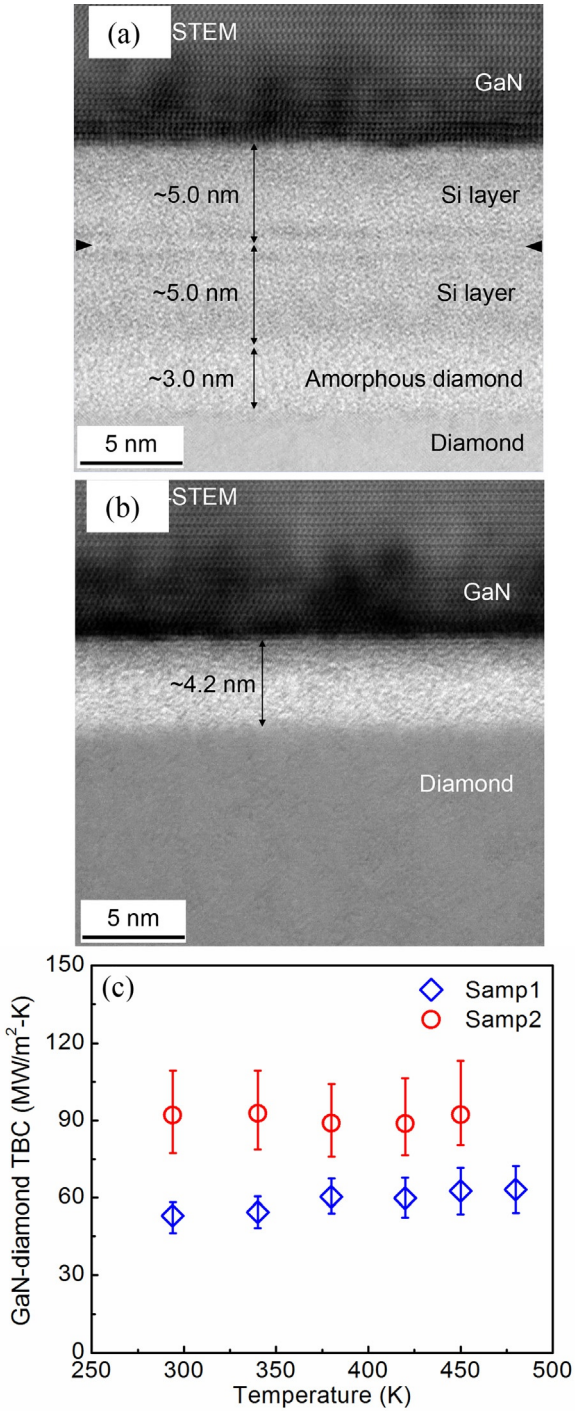


Fig. 16.9 (A) Cross-section scanning electron transmission microscopy image of GaN–diamond interfaces bonded with deposited Si (marked Samp1 in (C)). (B) Cross-section scanning electron transmission microscopy image of GaN–diamond interfaces bonded with Si-containing Ar ion beams (marked Samp2 in (C)). (C) The measured TBC of the two bonded GaN–diamond interfaces [31]. Reproduced with permission from Z. Cheng, F. Mu, L. Yates, T. Suga, S. Graham, Interfacial thermal conductance across room-temperature-bonded GaN/diamond interfaces for GaN-on-diamond devices, ACS Appl. Mater. Interfaces 12 (2020) 8376–8384.

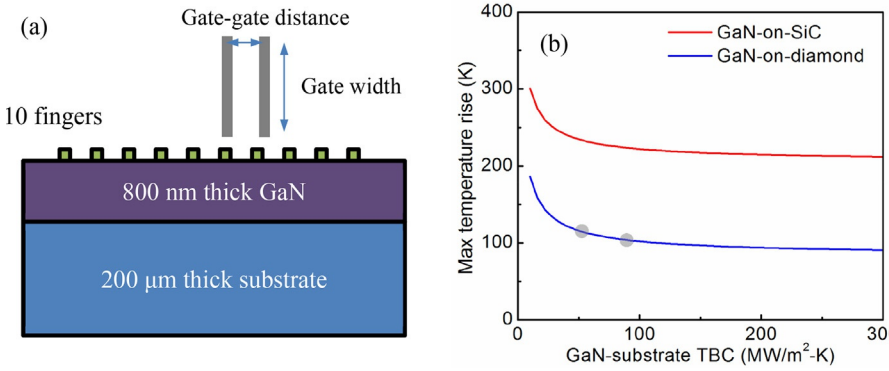


Fig. 16.10 (A) An 800-nm GaN device with 10 fingers with different substrates. The used power density is 10 W/mm [62]. (B) The calculated maximum device temperature with different substrates and GaN-substrate TBC [31].
Reproduced with permission from Z. Cheng, F. Mu, L. Yates, T. Suga, S. Graham, Interfacial thermal conductance across room-temperature-bonded GaN/diamond interfaces for GaN-on-diamond devices, ACS Appl. Mater. Interfaces 12 (2020) 8376–8384.

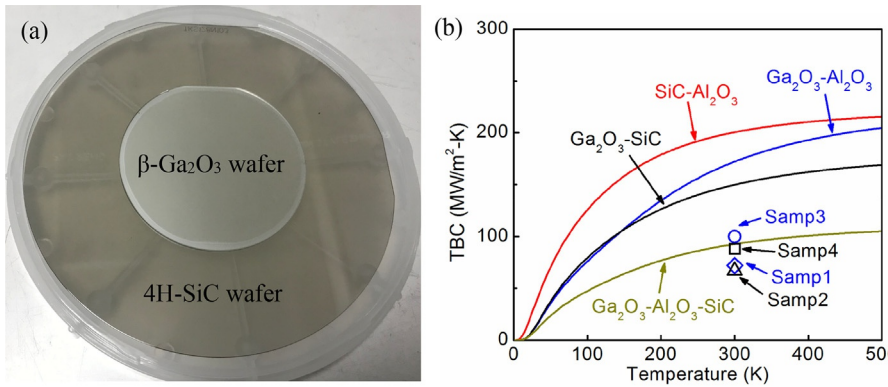


Fig. 16.11 (A) A 2-in. β-Ga_2O_3 wafer is bonded with a 4-in. 4H-SiC wafer by surface-activated bonding. (B) The measured β-Ga_2O_3–SiC TBC of four different samples. Samp1 and Samp2 have a 30-nm Al_2O_3 interlayer while Samp3 and Samp4 have a 10-nm Al_2O_3 interlayer. Samp1 and Samp3 are as-bonded while Samp2 and Samp4 are annealed at 1073 K after bonding.
Reproduced with permission from Z. Cheng, et al., Thermal transport across ion-cut monocrystalline β-Ga_2O_3 thin films and bonded β-Ga_2O_3–SiC interfaces, ACS Appl. Mater. Interfaces 12 (40) (2020) 44943–44951.

bonding of β-Ga_2O_3 wafers and 4H-SiC wafers have been demonstrated [64]. No voids or un-bonded area are observed.

Fig. 16.11B shows the measured TBC of the bonded β-Ga_2O_3–SiC interfaces compared to the calculated TBC of β-Ga_2O_3–SiC, β-Ga_2O_3–Al_2O_3, and Al_2O_3–SiC interfaces based on a Landauer Approach [57]. The phonon dispersion relations of

β-Ga_2O_3, Al_2O_3, and SiC are calculated by first-principles calculation. The phonon transmissions are calculated based on the Diffuse Mismatch Model [14,65]. Four samples were studied here. Samp1 and Samp2 have a 30-nm Al_2O_3 interlayer while Samp3 and Samp4 have a 10-nm Al_2O_3 interlayer. Samp1 and Samp3 are as-bonded while Samp2 and Samp4 are annealed at 1073 K after bonding. As shown in Fig. 16.11B, the measured TBC of interfaces with thicker Al_2O_3 interfacial layers have lower TBC because the extra Al_2O_3 brings in extra thermal resistances [57]. The annealing has negligible impact on the measured TBC partially due to the complicated interfacial structures. The TBC of the β-Ga_2O_3–Al_2O_3–SiC interface is obtained by adding the thermal resistance of the β-Ga_2O_3–Al_2O_3 interface and Al_2O_3–SiC interface. Overall, the measured TBC agree well with the calculated TBC, which confirms the excellent bonding quality of these interfaces. The TBC values are limited by the phonon properties of the materials instead of the bonding quality. Room-temperature surface-activated bonding technique opens the door for wafer-scale integration of semiconductors with high thermal conductance and low stress, which will impact applications of electronics thermal management, especially for power electronics. Furthermore, this bonding technique offers a way to create interfaces which cannot be grown before and provides additionally platform to study fundamental science of interfacial heat transport. The Ar ions create disorder at the interface and redistribute after annealing. This can be used as a tool to tune interfacial thermal conductance and understand the effects of localized vibrational modes and local structure on the interfacial heat transport.

Acknowledgments

The authors would like to acknowledge the funding support from U.S. Office of Naval Research under a MURI program (Grant No. N00014-18-1-2429) and U.S. Air Force Office of Scientific Research under a MURI program (Grant No. FA9550-18-1-0479).

References

[1] L. Yates, et al., Compound Semiconductor Integrated Circuit Symposium (CSICS), IEEE, 2016, pp. 1–4.
[2] H. Amano, et al., The 2018 GaN power electronics roadmap, J. Phys. D Appl. Phys. 51 (2018), 163001.
[3] J. Tsao, et al., Ultrawide—bandgap semiconductors: research opportunities and challenges, Adv. Electron. Mater. 4 (2018) 1600501.
[4] D. Francis, et al., Formation and characterization of 4-inch GaN-on-diamond substrates, Diamond Relat. Mater. 19 (2010) 229–233.
[5] B. Lu, D. Piedra, T. Palacios, The Eighth International Conference on Advanced Semiconductor Devices and Microsystems, IEEE, 2010, pp. 105–110.
[6] Z. Cheng, et al., Probing growth-induced anisotropic thermal transport in high-quality CVD diamond membranes by multi-frequency and multi-spot-size time-domain thermoreflectance, ACS Appl. Mater. Interfaces (2018).
[7] F. Faili, N.L. Palmer, S. Oh, D.J. Twitchen, ITherm, Orlando, Florida, US, 2017.
[8] C. Suckling, D. Nguyen, ARMMS Conference of RF and Microwave Society, 2012.

[9] D.G. Cahill, Analysis of heat flow in layered structures for time-domain thermoreflectance, Rev. Sci. Instrum. 75 (2004) 5119–5122.
[10] A.J. Schmidt, Pump-probe thermoreflectance, Annu. Rev. Heat Transf. 16 (2013).
[11] A.J. Schmidt, R. Cheaito, M. Chiesa, A frequency-domain thermoreflectance method for the characterization of thermal properties, Rev. Sci. Instrum. 80 (2009), 094901.
[12] J.T. Gaskins, et al., Thermal boundary conductance across heteroepitaxial ZnO/GaN interfaces: assessment of the phonon gas model, Nano Lett. 18 (2018) 7469–7477.
[13] Z. Cheng, et al., Thermal conductance across β-Ga2O3-diamond van der Waals heterogeneous interfaces, APL Mater. 7 (2019), 031118.
[14] Z. Cheng, et al., Thermal conductance across harmonic-matched epitaxial Al-sapphire heterointerfaces, Commun. Phys. 3 (2020) 1–8.
[15] X. Yang, T. Feng, J. Li, X. Ruan, Stronger role of four-phonon scattering than three-phonon scattering in thermal conductivity of III-V semiconductors at room temperature, Phys. Rev. B 100 (2019), 245203.
[16] L. Lindsay, D. Broido, T. Reinecke, Thermal conductivity and large isotope effect in GaN from first principles, Phys. Rev. Lett. 109 (2012), 095901.
[17] G.A. Slack, Nonmetallic crystals with high thermal conductivity, J. Phys. Chem. Solid 34 (1973) 321–335.
[18] H. Li, et al., GaN thermal transport limited by the interplay of dislocations and size effects, Phys. Rev. B 102 (2020), 014313.
[19] Q. Zheng, et al., Thermal conductivity of GaN, GaN 71, and SiC from 150 K to 850 K, Phys. Rev. Mater. 3 (2019), 014601.
[20] R. Rounds, et al., Thermal conductivity of GaN single crystals: influence of impurities incorporated in different growth processes, J. Appl. Phys. 124 (2018), 105106.
[21] P. Paskov, M. Slomski, J. Leach, J. Muth, T. Paskova, Effect of Si doping on the thermal conductivity of bulk GaN at elevated temperatures–theory and experiment, AIP Adv. 7 (2017), 095302.
[22] A. Jeżowski, et al., Thermal conductivity of heavily doped bulk crystals GaN: O. Free carriers contribution, Mater. Res. Express 2 (2015), 085902.
[23] R.B. Simon, J. Anaya, M. Kuball, Thermal conductivity of bulk GaN—effects of oxygen, magnesium doping, and strain field compensation, Appl. Phys. Lett. 105 (2014), 202105.
[24] H. Shibata, et al., High thermal conductivity of gallium nitride (GaN) crystals grown by HVPE process, Mater. Trans. 48 (2007) 2782–2786.
[25] C. Mion, J. Muth, E. Preble, D. Hanser, Accurate dependence of gallium nitride thermal conductivity on dislocation density, Appl. Phys. Lett. 89 (2006), 092123.
[26] A. Jeżowski, et al., Thermal conductivity of GaN crystals in 4.2–300 K range, Solid State Commun. 128 (2003) 69–73.
[27] G.A. Slack, L.J. Schowalter, D. Morelli, J.A. Freitas Jr., Some effects of oxygen impurities on AlN and GaN, J. Cryst. Growth 246 (2002) 287–298.
[28] E. Sichel, Thermal conductivity of GaN, 25–360 K, J. Phys. Chem. Solid 38 (1977) 330.
[29] T.E. Beechem, et al., Size dictated thermal conductivity of GaN, J. Appl. Phys. 120 (2016), 095104.
[30] F. Mu, et al., High thermal boundary conductance across bonded heterogeneous GaN-SiC interfaces, ACS Appl. Mater. Interfaces 11 (2019) 7.
[31] Z. Cheng, F. Mu, L. Yates, T. Suga, S. Graham, Interfacial thermal conductance across room-temperature-bonded GaN/diamond interfaces for GaN-on-diamond devices, ACS Appl. Mater. Interfaces 12 (2020) 8376–8384.
[32] E. Ziade, et al., Thickness dependent thermal conductivity of gallium nitride, Appl. Phys. Lett. 110 (2017), 031903.

[33] T.L. Bougher, et al., Thermal boundary resistance in GaN films measured by time domain thermoreflectance with robust Monte Carlo uncertainty estimation, Nanoscale Microscale Thermophys. Eng. 20 (2016) 22–32.
[34] J. Cho, et al., Phonon scattering in strained transition layers for GaN heteroepitaxy, Phys. Rev. B 89 (2014), 115301.
[35] A. Sarua, et al., Thermal boundary resistance between GaN and substrate in AlGaN/GaN electronic devices, IEEE Trans. Electron Devices 54 (2007) 3152–3158.
[36] C.-Y. Luo, H. Marchand, D. Clarke, S. DenBaars, Thermal conductivity of lateral epitaxial overgrown GaN films, Appl. Phys. Lett. 75 (1999) 4151–4153.
[37] D. Florescu, et al., Thermal conductivity of fully and partially coalesced lateral epitaxial overgrown GaN/sapphire (0001) by scanning thermal microscopy, Appl. Phys. Lett. 77 (2000) 1464–1466.
[38] V. Asnin, F.H. Pollak, J. Ramer, M. Schurman, I. Ferguson, High spatial resolution thermal conductivity of lateral epitaxial overgrown GaN/sapphire (0001) using a scanning thermal microscope, Appl. Phys. Lett. 75 (1999) 1240–1242.
[39] Z. Su, et al., Layer-by-layer thermal conductivities of the Group III nitride films in blue/green light emitting diodes, Appl. Phys. Lett. 100 (2012), 201106.
[40] D. Florescu, V. Asnin, F.H. Pollak, R. Molnar, C. Wood, High spatial resolution thermal conductivity and Raman spectroscopy investigation of hydride vapor phase epitaxy grown n-GaN/sapphire (0001): doping dependence, J. Appl. Phys. 88 (2000) 3295–3300.
[41] K. Park, C. Bayram, Impact of dislocations on the thermal conductivity of gallium nitride studied by time-domain thermoreflectance, J. Appl. Phys. 126 (2019), 185103.
[42] F. Mu, et al., High thermal boundary conductance across bonded heterogeneous GaN–SiC interfaces, ACS Appl. Mater. Interfaces 11 (2019) 33428–33434.
[43] E. Ziade, et al., Thermal transport through GaN–SiC interfaces from 300 to 600 K, Appl. Phys. Lett. 107 (2015), 091605.
[44] J. Cho, D. Francis, D.H. Altman, M. Asheghi, K.E. Goodson, Phonon conduction in GaN-diamond composite substrates, J. Appl. Phys. 121 (2017), 055105.
[45] A. Manoi, J.W. Pomeroy, N. Killat, M. Kuball, Benchmarking of thermal boundary resistance in AlGaN/GaN HEMTs on SiC substrates: implications of the nucleation layer microstructure, IEEE Electron Device Lett. 31 (2010) 1395–1397.
[46] J. Cho, et al., IEEE Compound Semiconductor Integrated Circuit Symposium (CSICS), IEEE, 2012, pp. 1–4.
[47] J.W. Pomeroy, M. Bernardoni, D. Dumka, D. Fanning, M. Kuball, Low thermal resistance GaN-on-diamond transistors characterized by three-dimensional Raman thermography mapping, Appl. Phys. Lett. 104 (2014), 083513.
[48] D. Dumka, et al., IEEE Compound Semiconductor Integrated Circuit Symposium (CSICS), IEEE, 2013, pp. 1–4.
[49] J. Cho, Y. Won, D. Francis, M. Asheghi, K.E. Goodson, IEEE Compound Semiconductor Integrated Circuit Symposium (CSICS), IEEE, 2014, pp. 1–4.
[50] H. Sun, et al., Reducing GaN-on-diamond interfacial thermal resistance for high power transistor applications, Appl. Phys. Lett. 106 (2015), 111906.
[51] J. Cho, et al., Improved thermal interfaces of GaN–diamond composite substrates for HEMT applications, IEEE Trans. Compon. Packag. Manuf. Technol. 3 (2013) 79–85.
[52] L. Yates, et al., Low thermal boundary resistance interfaces for GaN-on-diamond devices, ACS Appl. Mater. Interfaces 10 (2018) 24302–24309.
[53] Y. Zhou, et al., Barrier-layer optimization for enhanced GaN-on-diamond device cooling, ACS Appl. Mater. Interfaces 9 (2017) 34416–34422.

[54] J. Anaya, et al., Simultaneous determination of the lattice thermal conductivity and grain/grain thermal resistance in polycrystalline diamond, Acta Mater. (2017).
[55] R. Cheaito, et al., Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm), 2017 16th IEEE Intersociety Conference on, IEEE, 2017, pp. 706–710.
[56] Z. Cheng, et al., Significantly reduced thermal conductivity in beta-(Al0. 1Ga0. 9) 2O3/Ga2O3 superlattices, Appl. Phys. Lett. **115** (2019).
[57] Z. Cheng, et al., Thermal transport across ion-cut monocrystalline β-Ga_2O_3 thin films and bonded β-Ga_2O_3–SiC interfaces, ACS Appl. Mater. Interfaces 12 (40) (2020) 44943–44951.
[58] S. Mandal, et al., Thick, adherent diamond films on AlN with low thermal barrier resistance, ACS Appl. Mater. Interfaces 11 (2019) 40826–40834.
[59] A. Sood, et al., Anisotropic and inhomogeneous thermal conduction in suspended thin-film polycrystalline diamond, J. Appl. Phys. 119 (2016), 175103.
[60] S. Choi, E. Heller, D. Dorsey, R. Vetury, S. Graham, The impact of mechanical stress on the degradation of AlGaN/GaN high electron mobility transistors, J. Appl. Phys. 114 (2013), 164501.
[61] B. Kang, et al., Effect of external strain on the conductivity of AlGaN/GaN high-electron-mobility transistors, Appl. Phys. Lett. 83 (2003) 4845–4847.
[62] K.R. Bagnall, Y.S. Muzychka, E.N. Wang, Analytical solution for temperature rise in complex multilayer structures with discrete heat sources, IEEE Trans. Compon. Packag. Manuf. Technol. 4 (2014) 817–830.
[63] S. Pearton, et al., A review of Ga2O3 materials, processing, and devices, Appl. Phys. Rev. 5 (2018), 011301.
[64] Y. Xu, et al., Direct wafer bonding of Ga2O3–SiC at room temperature, Ceram. Int. 45 (2019) 6552–6555.
[65] E.T. Swartz, R.O. Pohl, Thermal boundary resistance, Rev. Mod. Phys. 61 (1989) 605.